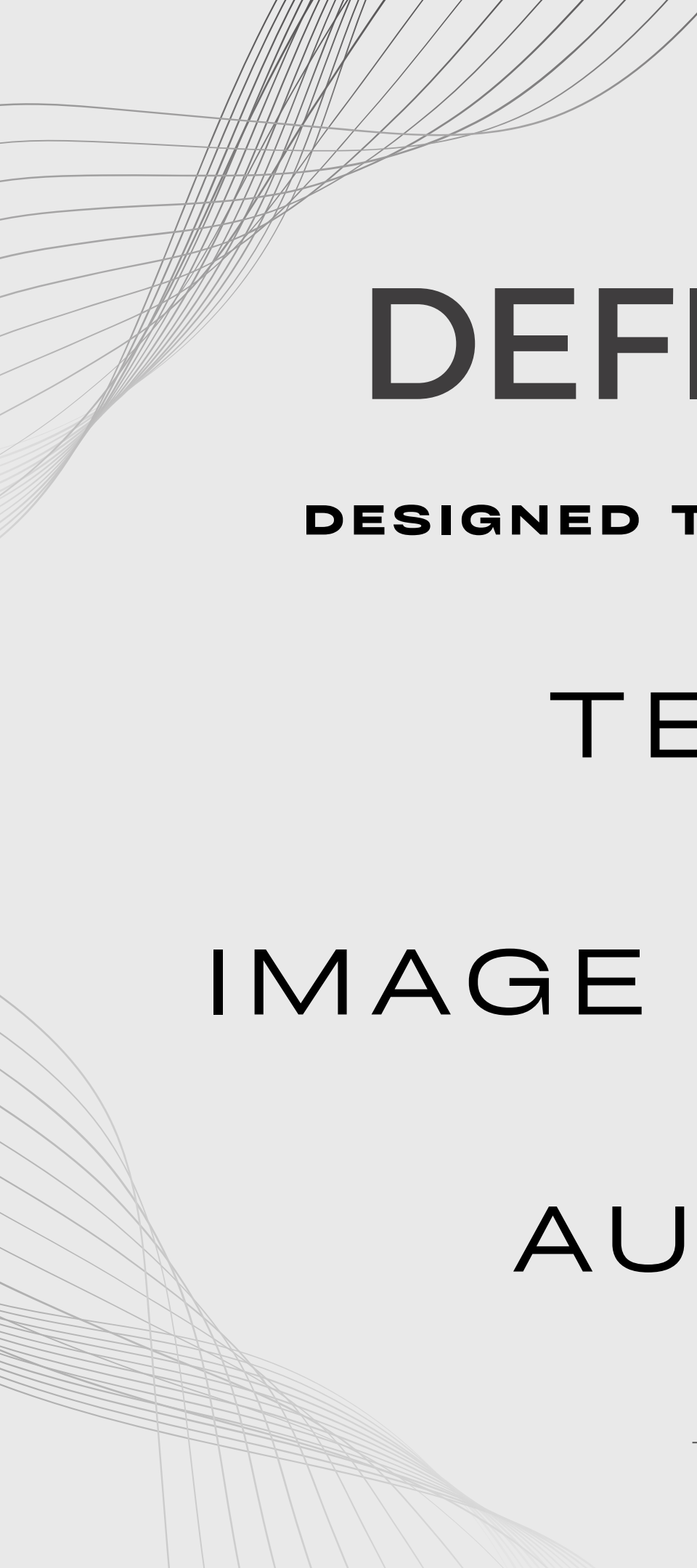


TRANSFORMERS

Rodinger Daniel | Zamiatala Daniel

Deep Learning Engineering



DEFINITION OF MODEL

DESIGNED TO HANDLE SEQUENTIAL DATA USING ATTENTION

TEXT

TRANSLATION,
SUMMARIZATION,
SENTIMENT ANALYSIS,
GENAI (CODE, TEXT)

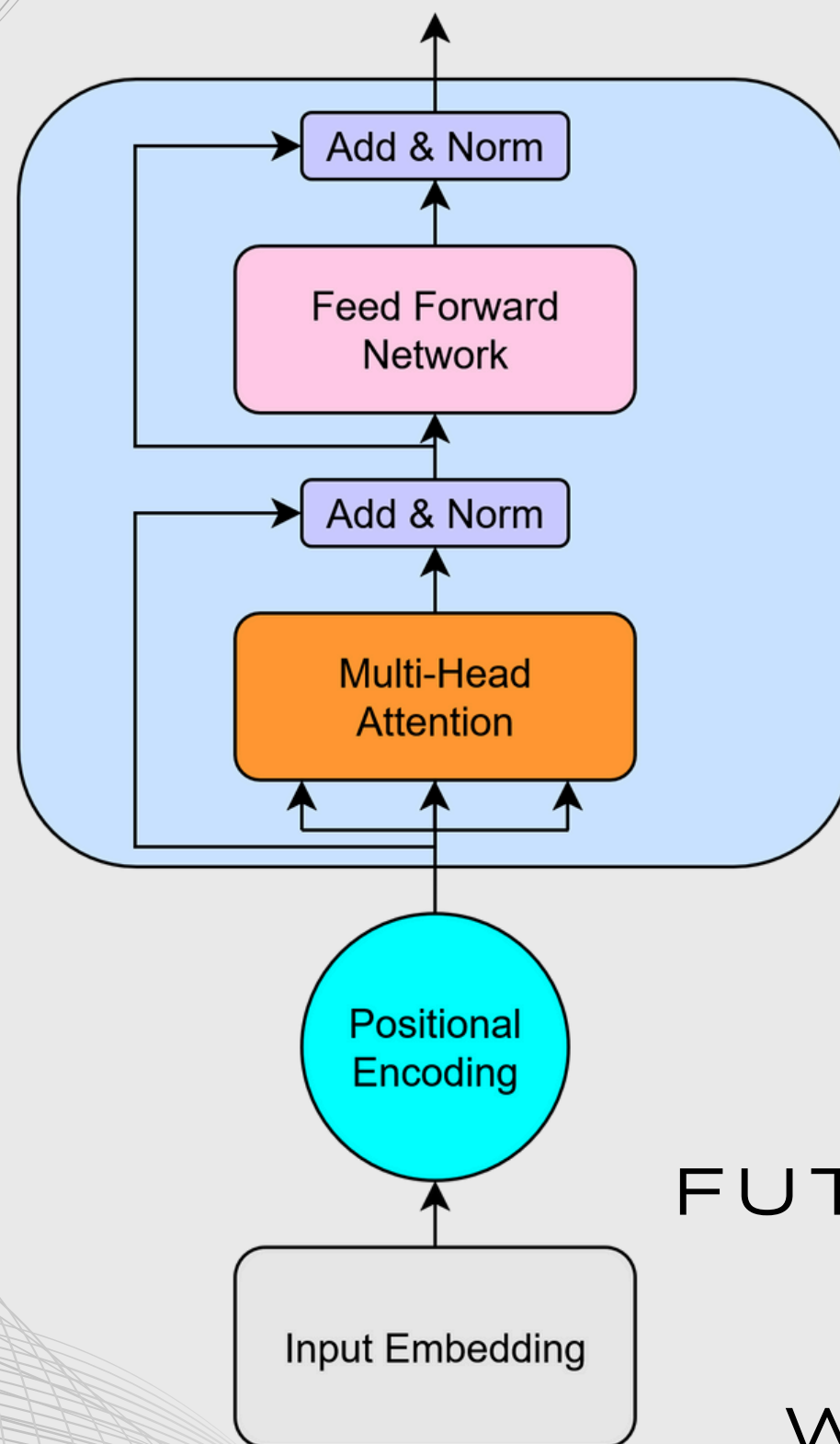
IMAGE / VIDEO

OBJECT DETECTION,
IMAGE CLASSIFICATION

AUDIO

SPEECH RECOGNITION,
TRANSCRIPTION

ARCHITECTURE



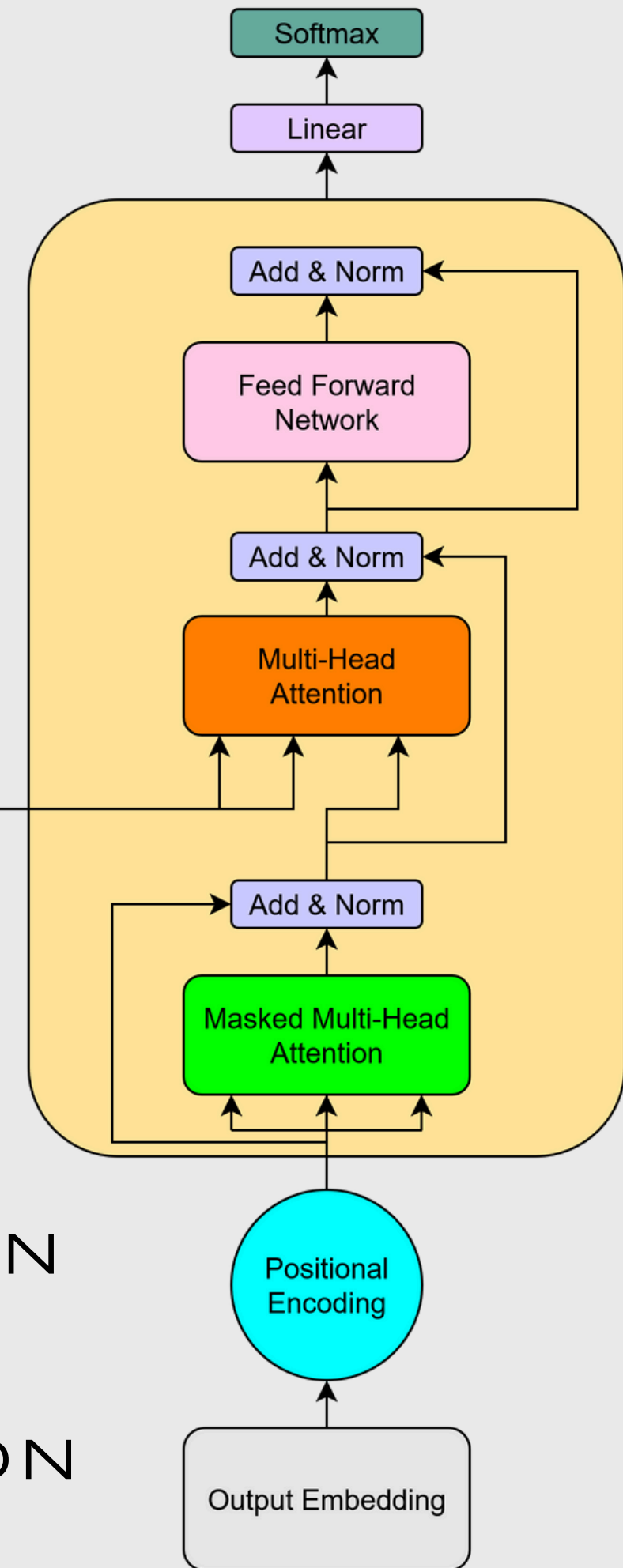
SOFTMAX
LOGITS → PROBABILITY

LINEAR
PREDICTS NEXT TOKEN

ATTENTION
HOW MUCH EACH WORD
ATTEND TO OTHERS

MASKED ATTENTION
FUTURE TOKENS ARE HIDDEN

POSITIONAL ENCODING
WORD ORDER INFORMATION





NOVELTY

PARALLELIZATION (FASTER THAN RNN/LSTM)

LONG RANGE CONTEXT UNDERSTANDING

SCALABILITY

MULTIMODAL INTEGRATION

PERFORMANCE AND ACCESSIBILITY

SOTA ACROSS ALL NLP BENCHMARKS

STRONG ZERO-SHOT PERFORMANCE

MANY OPEN-SOURCE MODELS VIA HF





DISADVANTAGES

HIGH COMPUTATIONAL COST FOR TRAINING / INFERENCE

LARGE LLMS PRONE TO HALLUCINATION AND BIAS

REQUIRE LARGE DATASETS